Social Event Mapper

Cloud Computing Term Project - Phase 1

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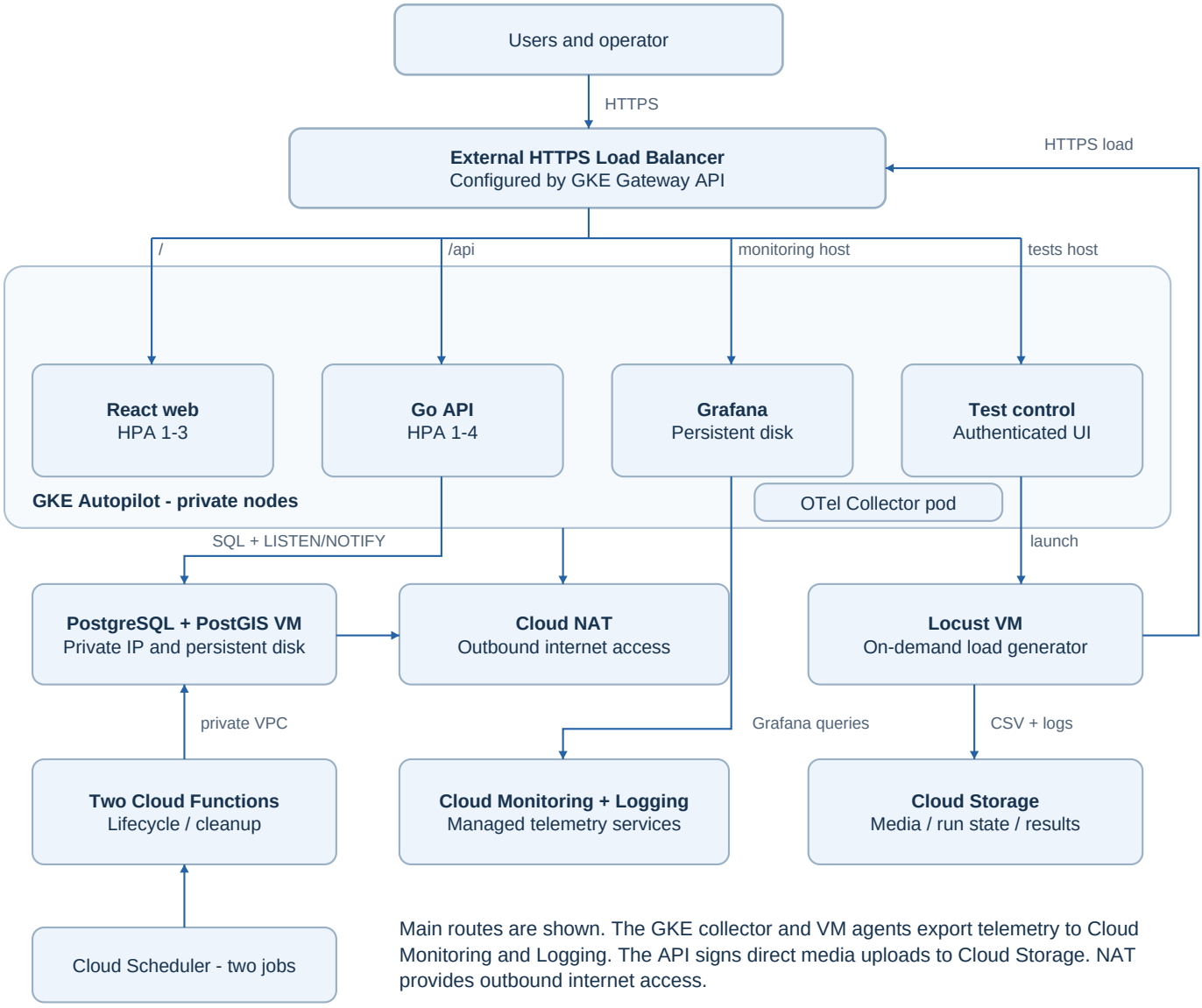
Partner None (individual project)

Application code github.com/kaanunsel/social-event-mapper

Tentative cloud architecture



Main routes are shown. The GKE collector and VM agents export telemetry to Cloud Monitoring and Logging. The API signs direct media uploads to Cloud Storage. NAT provides outbound internet access.

GKE serves the React application and Go API with independent autoscaling. The external load balancer routes by hostname and path and removes the /api prefix. PostgreSQL provides spatial queries and LISTEN/NOTIFY for notification delivery across API pods.

Private nodes and the database have no public IP. Cloud NAT supports package downloads and enabled external integrations. Cloud Scheduler invokes separate functions for minute-level event updates and daily notification cleanup. Terraform will provision the infrastructure.

Evaluation and estimated cost

Grafana will show component logs, resource metrics and load-test results. An authenticated test control service will launch Locust on a temporary VM, store CSV outputs and logs in Cloud Storage, and remove the VM after each run. Application images will use a separate Cloud Storage bucket.

Locust will compare fixed and autoscaled deployments using event discovery, detail and participation workloads. Measurements will include p95 latency, requests per second, errors, CPU, memory and replica count.

Planning assumptions

Iowa (us-central1), USD on-demand rates, 168 or 720 deployment hours. Five baseline pods request 1.5 vCPU and 3.5 GiB RAM in total. Tests run for 8 or 24 hours at the proposed HPA maxima. Storage includes 30 GiB for PostgreSQL, 10 GiB for Grafana and 10 GiB of application media.

Component	7 days	30 days
Web and API baseline pods	$6.89	$29.55
Grafana, test control and OTel Collector	$7.33	$31.42
Grafana disk and application media	$0.28	$1.18
Additional pods during load tests	$0.88	$2.63
Locust VM, disk and outbound IPv4	$0.32	$0.96
PostGIS VM and disk	$6.32	$27.08
HTTPS load balancer	$4.20	$18.00
Cloud NAT and NAT IPv4 allowance	$1.78	$7.63
Two functions and Scheduler jobs*	$0.00	$0.00
Traffic, telemetry, builds and storage allowance	$5.00	$20.00
GKE management before credit	$16.80	$72.00
Total before GKE management credit	**$49.80**	**$210.45**
Total with available management credit	**$33.00**	**$138.45**

*Assumes available free allowances, zero idle function instances and an average of three billable seconds per invocation. GKE management credit is up to $74.40/month per billing account. NAT assumes four private VM/node equivalents. Totals use unrounded amounts.

These are alternative deployment windows within the $300 project budget. All project usage accumulates against the remaining trial credit. An optional read replica adds about $6.32 / $27.08; a second may follow if needed. Replication traffic and failover require further budgeting.

Pricing checked 12 November 2026. Sources: GKE, Compute Engine, disks, load balancing, NAT, Storage, functions, Scheduler.